

Validity

Introduction

A scale possesses validity when it actually measures what it claims to measure. In other words, a scale is said to be valid if it measures what is expected to measure.

Interpretation of test scores ultimately involves predictions about a subject's behavior in a specified situation. If a test is an accurate predictor, it is said to have good validity.

Types of validity:

1. Content Validity

Content validity is the representativeness or adequacy of the unit selected of the content such as the substances, the matter, the topic of the measuring instrument etc. Content Validity is also known as logical validity. A test has content validity if the sample of items in the test is representative of all the relevant items that might have been used.

Let, U be the universe of the item; S is the subset of U i.e., $S \subset U$ and x be the item such that x is the element in U then it must be an element of S , i.e., $x \in U \Rightarrow x \in S$

The standardized achievement test is used for the content validity measure.

Content validity is most often addressed in academic and vocational testing, where test items need to reflect the knowledge actually required for a given topic area or job skill. In clinical settings, content validity refers to the correspondence between test items and the symptom content of a syndrome.

2. Criterion-related Validity

A common approach, called criterion related validity is to correlate measures with a criterion measure known to be valid. For example, an art-aptitude test has predictive validity if high scores are achieved by those who later do well in art school. The concurrent validity of a new intelligence test may be demonstrated if its scores correlate closely with those of an already well-established test.

The criterion related validity is based on the four decisive factors: (i) An external criterion (ii) regular and future behavior (iii) Logical analysis and (iv) Empirical method.

- It is used to predict the criterion on the basis of some measure.
- It reflects the success of measures for empirical estimating process

In this type of validity, the proposed criterion must possess the following quality:

- Freedom from bias (criterion should be given each subject/matter an equal opportunity to score)
- Reliable (criterion should be stable and reproducible)
- Relevance (criterion should be defined in terms of proper measure). Availability (information specified by the criterion must be available)

The Criterion- related Validity broadly classified as: (a) Predictive validity and (b) Concurrent Validity

(a) Predictive validity:

It refers to the usefulness of the test in prediction some future performances on the criterion. It is concerned with how well the scale can forecast a future criterion.

(b) Concurrent Validity:

It is concerned with the performances that how it can describe a present criterion. Concurrent Validity is demonstrated where a test correlates well with a measure that has previously been validated. For example, if a test measuring job satisfaction gives similar results to those gathered using a job satisfaction which has been validated in past investigations the new measurement has concurrent validity.

The validity coefficient is measured in terms of the correlation coefficient (r) of the scores of the different tests. We say, for $0.9 \leq r \leq 1$ there is very high validity; for $0.8 \leq r \leq 0.9$ a high validity; for $0.6 \leq r \leq 0.8$ a satisfactory validity; for $0.4 \leq r \leq 0.6$ a moderate validity; for $0.0 \leq r \leq 0.4$ a poor validity; and for $r < 0.0$ a negative validity

3. Construct Validity

It is the validity, which is most complex and abstract because of the complexity of the social parameter. It is based on psychological trait and quality. Construct validity is generally determined by investigating what psychological traits or qualities a test measures; that is, by demonstrating that certain patterns of human behavior account to some degree for performance on the test.

Construct validity can be evaluated by statistical methods that show whether or not a common factor can be shown to exist underlying several measurements using different observable indicators. For determining construct validity, we associate a set of other proposition with the result received from the use of our measuring instrument. If measurements on our devised scale correlated (associated) in a predicted way with the other propositions, we can conclude that there is construct validity.

Scaling

Scores and Scales

Scores

The number of points somebody gets for correct answers in a test is said to be scores. In other words the value of parameter in the observed phenomenon is termed as score. From an experiment or from a test what we obtained as the observation is called the raw score. The raw score is the simple numerical count of responses such as the number of correct answers on an intelligence test. The usefulness of the raw score is limited however, because it does not convey how well someone does in comparison with others taking the same test. Suppose, one attempts to answer 20 IQs having 1 point each and answered 16 correctly then the raw score is 16.

Scale

Scale is a predefined sequence of scores in ascending values that can map an item to it. Scale is a set of all the different levels of symbols or numerals or something so constructed, from the lowest to highest, that these can be assigned by rule to objects or to items or to the individuals or to their behavior to whom it is applied. In general concept, scale is also known as a quantifying appliance used to indicate the systematized numerals of the measuring instrument.

Scaling of the Scores

From a set of scores of the test we can construct a sequence of levels of the values that can be used as an extent for the test of that phenomena, this method of leveling is said to be scaling. For the purpose of scaling it is always desired to make the scores in an array. After that the scores are converted to the percentile points and then to a scale of required form.

Scores which are modified/improved/developed from the raw scores are called derived scores. From the derived scores of the same form in sequence can be used to create a continuous structure of the numerals which is the scale required to be constructed.

There are different types of derived scores widely used in the measurement of the phenomenon or the attitudes. The percentile scores, s-score (z-score) and T-score are such derived scores and are used to compare the strength and credibility of the measures. From these scores we can construct the standard scales namely percentile scale, sigma scale (z-scale) and T scale, respectively by arranging the scores lowest to highest.

Scales used in Social and Physical sciences

The main scales used in the measure of social/physical characteristics are:

- (i) Point scale
- (ii) Social distance scale

- (iii) Rating scale
- (iv) Ranking scale
- (v) Thurstone scale

(i) Point scale

In this type of scale words or situations representing the criteria are selected and one point (marks or number) is given for each criteria. Attitude of a person can be determined by the use of all the three following methods effectively.

Method 1: The respondent is asked to tick one that is representing or favorable to him/her. The scores are counted and result is derived.

Method 2: In the second method the respondent is asked to cross the one point or situation which is not favorable to him/her. A point is given to each and every word that has not been crossed. The attitude of the respondent is then determined by counting no. of points.

Method 3: In the third method of point scale the respondent is asked to cross, on which points he/she is agree or not.

(ii) Social distance scale

The social distance scale is developed by E. S. Borgardus to measure the social distances (it is commonly known as Borgardus scale). The social distance may be defined as the proximity and favoritism; for example the cultural distance from one race to the other, custom from one ethnic group to the other etc. To measure a person's (respondent's) attitude how far from the given cause situation the Borgardus scale can be used. Borgardus developed a scale to measure the nearness of liking between two social groups using several items or statements which show the varying relationship of social distance of Americans with other races as English, Korean, Swedish and poles.

(iii) Rating scale

When the character to be measured is not dichotomous in this case the rating scale is used. Rating scale consists of a set of figures that can match to the individual or items to be measured. The response or the opinions of the respondent's attitude is rated in three to six points in continuum (range). The intensity of the attitude is measured by using equal or unequal type intervals. An example of three point rating scale is:

Very goodsatisfactory.....poor

A five points rating scale is

Strongly-Agree.....AgreeNot-Decided.Dis-agree.....Strongly-Disagree

(iv) ***Ranking scale***

The ranking scale is similar to rating scale applied to a set of objects or individuals with the preference or liking. In this scale the situations are placed in such a way that, everybody who inspects it knows that one likes the one better than the other. Ranking scales is determined in comparison to a few cases known as stimuli. The item obtaining first preference scores 1, the second as 2, third as 3 and so on. 'The smaller the score the greater the preferences' is the principle of ranking scale.

(v) ***Thurstone scale***

American psychologist Louis L. Thurstone proposed that intelligence was not one general factor but a small set of independent factors of equal importance. He called these factors primary mental abilities. To identify these abilities, he developed a plan to conduct study amongst 250 college students, identified factors and developed a scale of measuring aptitude using factor analysis. In educational and psychological experiments, it is used as a main type of scale used to measure the attitude. The statements are collected and arranged in continuum from most favorable to least favorable with neutral point (zero). It is one type of point scale having neutrality point at the central location.

Most- Favored.....neutralLeast— Favored